

Obesity-associated mechanisms of hepatocarcinogenesis.

Obesity has been recognized as a key component of the metabolic syndrome, a cluster of risk factors associated with diabetes and cardiovascular morbidity. In addition, obesity has been linked to higher frequency of cancers in a variety of tissues including the liver. Liver cancer most often occurs as hepatocellular carcinoma (HCC) complicating cirrhosis due to chronic viral infection or toxic injury and remains the third leading cause of cancer death in the world. However, HCC is increasingly diagnosed among individuals with obesity and related disorders. As these metabolic conditions have become globally prevalent, they coexist with well-established risk factors of HCC and create a unique challenge for the liver as a chronically diseased organ. Obesity-associated HCC has recently been attributed to molecular mechanisms such as chronic inflammation due to adipose tissue remodeling and pro-inflammatory adipokine secretion, ectopic lipid accumulation and lipotoxicity, altered gut microbiota, and disrupted senescence in stellate cells, as well as insulin resistance leading to increased levels of insulin and insulin-like growth factors. These mechanisms synergize with those occurring in chronic liver disease resulting from other etiologies and accelerate the development of HCC before or after the onset of cirrhosis. Increasingly common interactions between oncogenic pathways linked to obesity and chronic liver disease may explain why HCC is one of the few malignancies with rising incidence in developed countries. Better understanding of this complex process will improve our strategies of cancer prevention, prediction, and surveillance.

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